

JEE MAIN REVISION MASTER: MATHEMATICAL REASONING OUT OF SYLLABUS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 121 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Mathematical Reasoning — Standard Calculations & Problem Archetypes	65 PYQs	Quantitative Analysis
Very High	Mathematical Reasoning — Applied Cases, Reactions & Traps	34 PYQs	Applied Topics
Very High	Mathematical Reasoning — Core Principles & Governing Laws	24 PYQs	Core Principle

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Mathematical Reasoning — Standard Calculations & Problem Archetypes	$\neg p \vee \neg(\neg p \vee \neg q), \quad B = p \wedge r, \quad p = \text{you are not born in India.}$
Mathematical Reasoning — Applied Cases, Reactions & Traps	$\therefore \neg[p \vee (\neg(p \wedge q))] = \neg p \wedge (p \wedge q), \quad (p \rightarrow q) = \neg(p \rightarrow q) = p \wedge \neg q, \quad S_2: p \rightarrow (q \vee r)$
Mathematical Reasoning — Core Principles & Governing Laws	$(P \wedge \neg R) \Leftrightarrow \neg Q, \quad \text{If } \Delta v = v, \nabla v = v, \quad (f \vee A) = A$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Mathematical Reasoning — Standard Calculations & Problem Archetypes (Very High Yield — 65 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Mathematical Reasoning.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2023 Morning Shift] Q#1 (MCQ): "Negation of $p \wedge (q \wedge \neg(p \wedge q))$ is :..." [Ans: N/A]
- [2023 Evening Shift] Q#1 (MCQ): "The converse of $(\neg p) \wedge q \Rightarrow r$ is..." [Ans: N/A]
- [2023 Morning Shift] Q#1 (MCQ): "Negation of $(p \Rightarrow q) \Rightarrow (q \Rightarrow p)$ is :..." [Ans: N/A]

3.2 Mathematical Reasoning — Applied Cases, Reactions & Traps (Very High Yield — 34 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Mathematical Reasoning.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2023 Evening Shift] Q#2 (MCQ): "The statement $(p \wedge (\neg q)) \vee ((\neg p) \wedge q) \vee ((\neg p) \wedge (\neg q))$ is equivalent to _____." [Ans: N/A]
- [2023 Evening Shift] Q#2 (MCQ): "The statement $\neg[p \vee (\neg(p \wedge q))]$ is equivalent to :..." [Ans: N/A]
- [2023 Evening Shift] Q#2 (MCQ): "Among the statements (S1) : $(p \Rightarrow q) \vee ((\neg p) \wedge q)$ is a tautology (S2) : $(q \Rightarrow p) \Rightarrow ((\neg p) \wedge q)$ is a c..." [Ans: N/A]

3.3 Mathematical Reasoning — Core Principles & Governing Laws (Very High Yield — 24 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Mathematical Reasoning.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Mathematical Reasoning.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Mathematical Reasoning PYQs.

Representative Source PYQs:

- [2023 Morning Shift] Q#1 (MCQ): "Negation of $p \wedge (q \wedge \neg(p \wedge q))$ is :..." [Ans: N/A]
- [2023 Morning Shift] Q#3 (MCQ): "The negation of the statement $((A \wedge (B \vee C)) \Rightarrow (A \vee B)) \Rightarrow A$ is..." [Ans: N/A]
- [2023 Morning Shift] Q#3 (MCQ): "The negation of the statement $(p \vee q) \wedge (q \vee (\neg r))$ is :..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2023 (Online) 15th April Morning Shift (Q#1)

Negation of $p \wedge (q \wedge \neg(p \wedge q))$ is :

- (A) $(\neg(p \wedge q)) \wedge q$
- (B) $(\neg(p \wedge q)) \vee p$
- (C) $p \vee q$
- (D) $\neg(p \vee q)$

Correct Answer: None

Solution Analysis: The given statement is: $p \wedge (q \wedge \neg(p \wedge q))$ We want to find the negation of this statement. To do that, we negate the whole expression: $\neg(p \wedge (q \wedge \neg(p \wedge q)))$ Now, we will use De Morgan's laws to simplify the expression. De Morgan's laws state: $\neg(A \wedge B) \equiv \neg A \vee \neg B$ $\neg(A \vee B) \equiv \neg A \wedge \neg B$ Applying the first De Morgan's law to the outer conjunction of the negated expression: $\neg p \vee \neg(q \wedge \neg(p \wedge q))$ Now we need to address the inner conjunction $(q \wedge \neg(p \wedge q))$. To simplify this part of the expression, we again apply the first De Morg

Benchmark Problem 2: JEE Main 2023 (Online) 11th April Evening Shift (Q#1)

The converse of $((\sim p) \wedge q) \Rightarrow r$ is

- (A) $((\sim p) \vee q) \Rightarrow r$
- (B) $(\sim r) \Rightarrow p \wedge q$
- (C) $(p \vee (\sim q)) \Rightarrow (\sim r)$
- (D) $(\sim r) \Rightarrow ((\sim p) \wedge q)$

Correct Answer: None

Solution Analysis: Converse of $((\sim p) \wedge q) \Rightarrow r \equiv r \Rightarrow (\sim p \wedge q) \equiv \sim r \vee (\sim p \wedge q) \equiv \sim r \vee (p \vee \sim q) \equiv (p \vee \sim q) \Rightarrow \sim r$

Benchmark Problem 3: JEE Main 2023 (Online) 13th April Evening Shift (Q#2)

The statement $(p \wedge (\sim q)) \vee ((\sim p) \wedge q) \vee ((\sim p) \wedge (\sim q))$ is equivalent to _____.

- (A) $(\sim p) \vee (\sim q)$
- (B) $p \vee (\sim q)$
- (C) $p \vee q$
- (D) $(\sim p) \vee q$

Correct Answer: None

Solution Analysis: $p \ q \ \sim q \ \sim p \ p \wedge \sim q \ \sim p \wedge q \ \sim p \wedge \sim q \ T \ T \ F \ F \ F \ F \ T \ F \ T \ F \ T \ F \ F \ F \ T \ F \ T \ F \ F \ F \ T \ T \ F \ F \ T$
 Using the truth table, we can see that the given statement is equivalent to $(\sim p) \vee (\sim q)$. Alternate Method : The given statement $(p \wedge (\sim q)) \vee ((\sim p) \wedge q) \vee ((\sim p) \wedge (\sim q))$ can be simplified using the following steps:
$$\begin{aligned} & (p \wedge (\sim q)) \vee ((\sim p) \wedge q) \vee ((\sim p) \wedge (\sim q)) \\ & \equiv (p \wedge (\sim q)) \vee ((\sim p) \wedge (q \vee (\sim q))) \\ & \equiv (p \wedge (\sim q)) \vee ((\sim p) \wedge T) \\ & \equiv (p \wedge (\sim q)) \vee (\sim p) \end{aligned}$$